

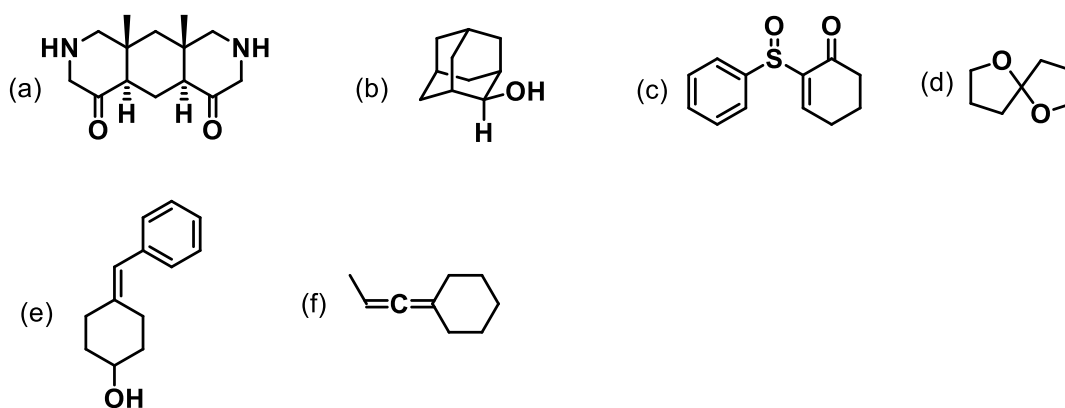
## CHM102 Basic Organic Chemistry

### Stereochemistry

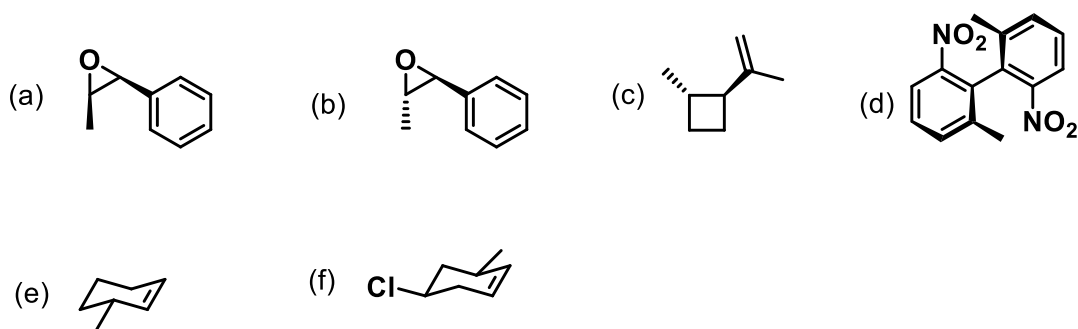
**Q1.** Draw all possible stereoisomers for each of the following compounds. State if no stereoisomers are possible.

- a. 1-bromo-2-chlorocyclohexane
- b. 1,2-dichlorocyclohexane
- c. 1,2-dimethylcyclopropane

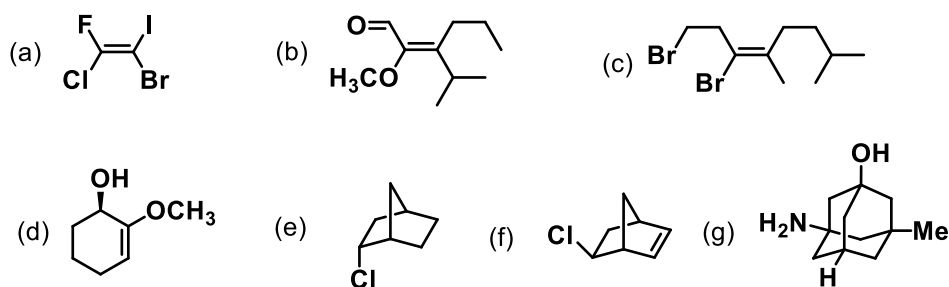
**Q2.** Identify whether the following molecules are chiral. Also, specify whether the compound is enantiomer or diastereomer. If not, classify the compound as achiral, meso compound.



**Q3.** Draw enantiomers of the following stereoisomers.



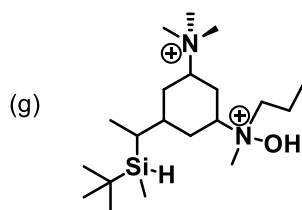
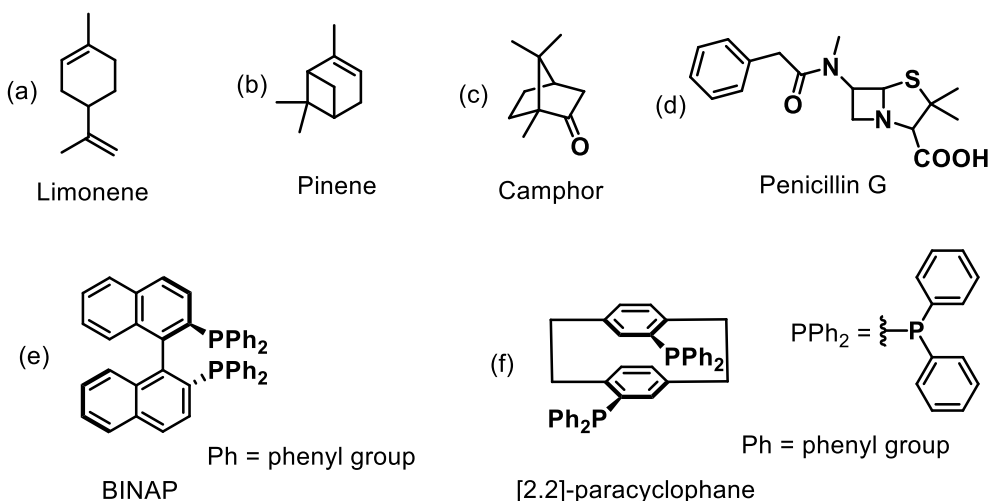
**Q4.** Name each of the following compounds using *R*, *S* and *E*, *Z* designations, wherever appropriate:



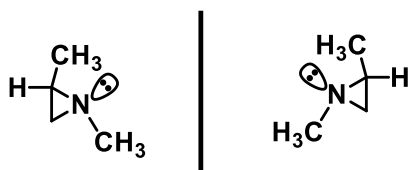
**Q5.** Which of the following compounds have an achiral stereoisomer?

- |                                     |                              |
|-------------------------------------|------------------------------|
| (a) 2,3-dichlorobutane              | (f) 2,4-dibromopentane       |
| (b) 2,3-dichloropentane             | (g) 2,3-dibromopentane       |
| (c) 2,3-dichloro-2,3-dimethylbutane | (h) 1,4-dimethylcyclohexane  |
| (d) 1,3-dichlorocyclopentane        | (i) 1,2-dimethylcyclopentane |
| (e) 1,3-dibromocyclobutane          | (j) 1,2-dimethylcyclobutane  |

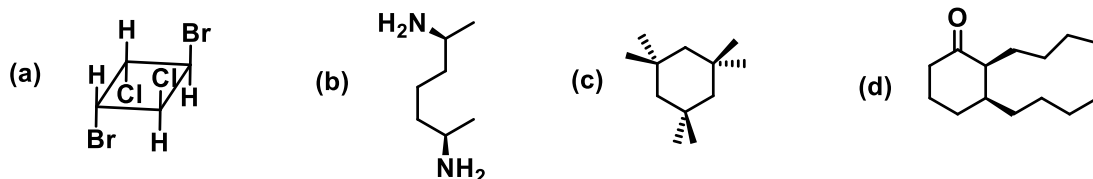
**Q6.** How many chiral centers are available in the following molecules? Indicate the chiral centers and/or type of chirality. Also, how many stereoisomers are possible for them?



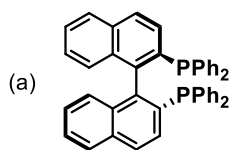
**Q7.** The enantiomers of 1,2-dimethylaziridine can be separated even though one of the "groups" attached to nitrogen is a lone pair. Explain.



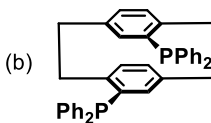
**Q8.** Whether the following molecule are chiral or optically active?



**Q9.** Identify *P* and *M* configurations (also *R* and *S* configurations) for the following compounds:

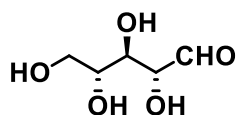


Ph = phenyl group



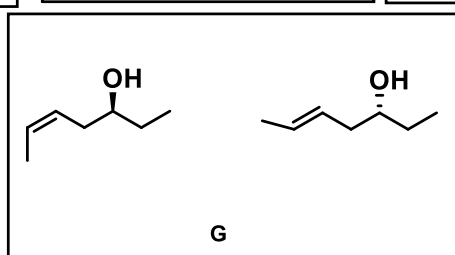
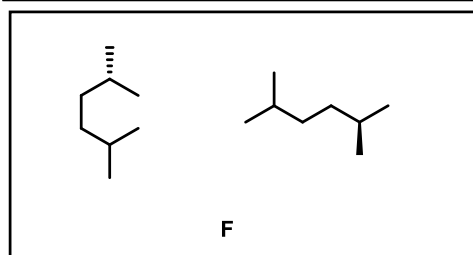
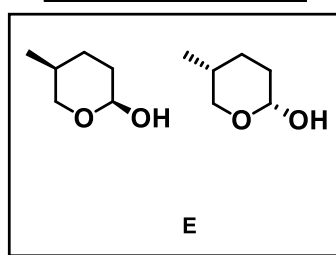
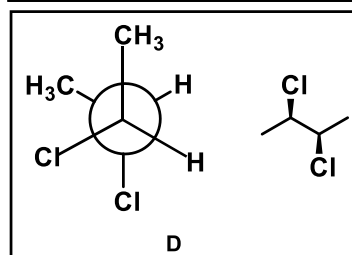
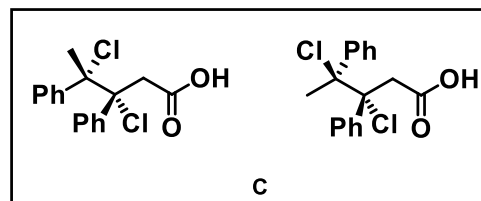
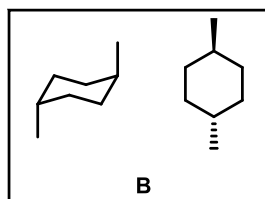
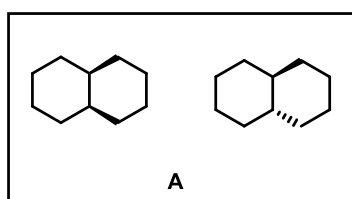
Ph = phenyl group

**Q10.** Draw the possible diastereomers and their enantiomeric pairs for the following molecule.

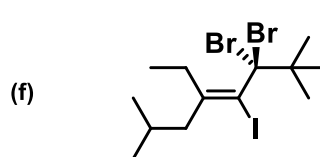
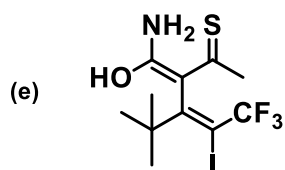
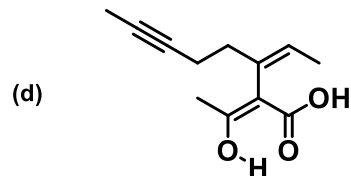
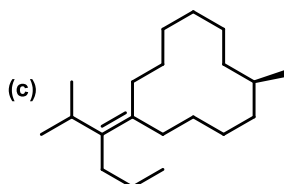
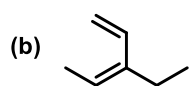
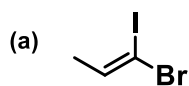


(2*R*,3*R*,4*R*)-2,3,4,5-tetrahydroxypentanal

**Q11.** The relationship between the following pair of molecules among the possibilities “Identical/Constitutional isomers/Enantiomers/Diastereomers” is

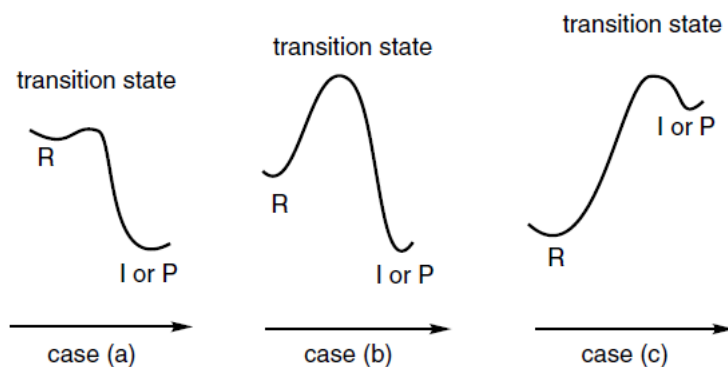


**Q12.** Assign absolute configurations (*E/Z*) to the following compounds. For poly-olefins, assignments should be made at each olefin. If it is not possible to assign, indicate the reason.



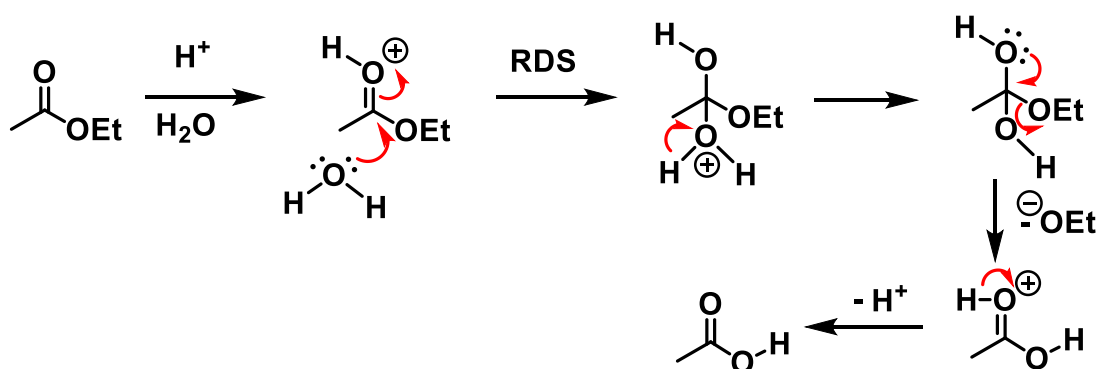
### Energy profile diagrams and reactivity aspects

**Q13.** Account for the structure of the transition state for the following cases based on the energy profiles of the reactions (R = Reactant; I = Intermediate; P = Product):



**Q14.** For the following reaction, draw an energy profile diagram (based on the available data) and account for the enthalpy/entropy control in the rate determining step.

### Acid-catalyzed hydrolysis of ethyl acetate in water

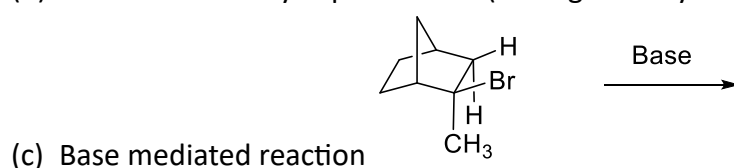


**Q13.** Construct energy profile diagram based on the known reaction mechanism for the following: (Also, indicate the rate determining step in each reaction)

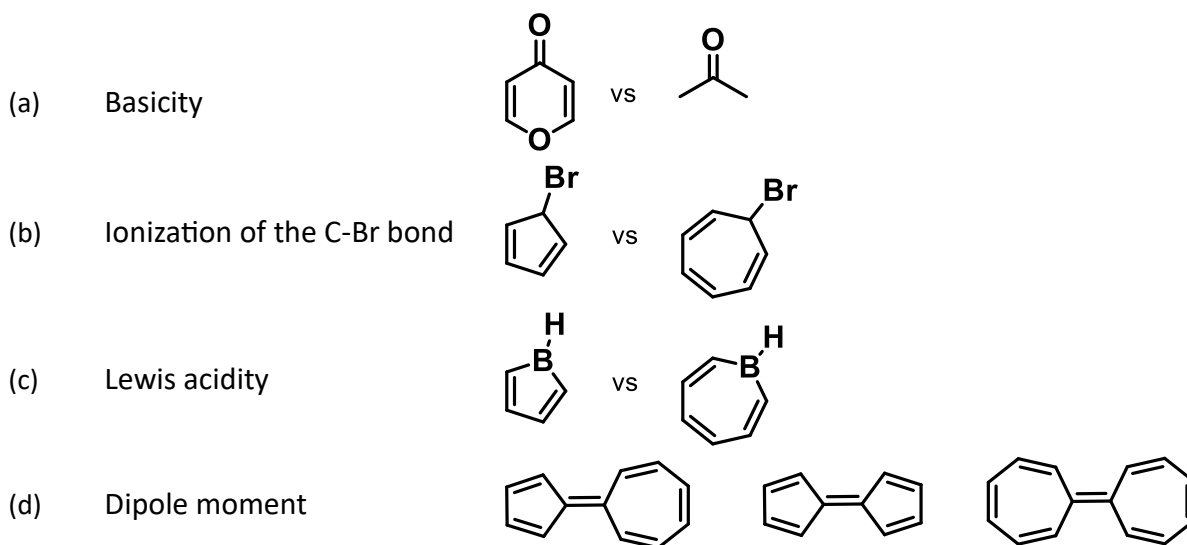
- Aromatic nucleophilic substitution reaction ( $\text{S}_{\text{N}}\text{Ar}$  (addition-elimination reaction))
- $\text{E}_1$ ,  $\text{E}_2$  and  $\text{E}_{1\text{CB}}$  reactions

**Q14.** Indicate the dominant factor among enthalpy and entropy in controlling the following reactions:

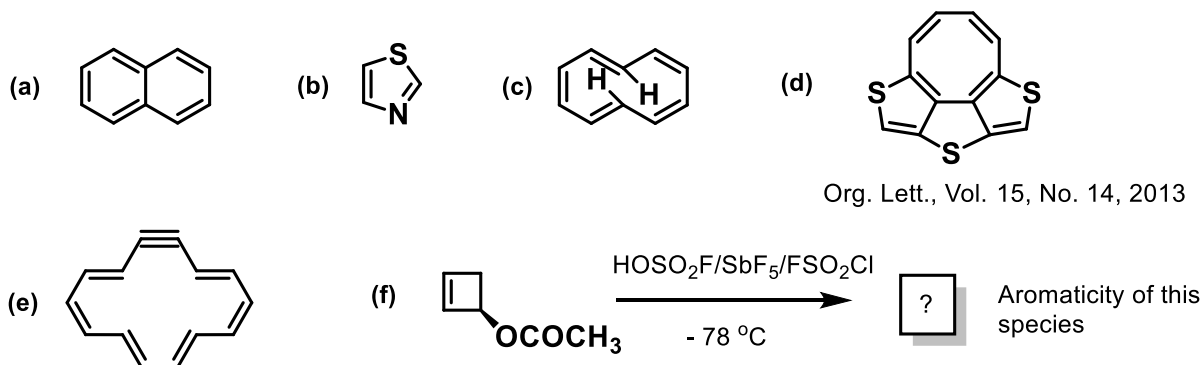
- Thermal decomposition of benzoyl peroxide
- Dimerization of cyclopentadiene (through 4+2 cycloaddition / Diels-Alder reaction)



**Q14.** Identify the molecule among each pair exhibiting the dominant property indicated for each case and indicate the reason.



**Q14.** Classify the following molecules according to their aromaticity characteristics:



**Q16.** Draw the molecular orbitals (MO) diagrams of the following with appropriate energetic ordering:

- (a) Cyclopentadienyl cation
- (b) Cyclopentadienyl radical
- (c) Cyclopentadienyl anion